

ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 21-05-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	4	4	3	4	1	3	2	3	4	1	3	1	4	3	3	4	3	3	4	4	3	3	2	3	3	2	1	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	2	3	2	3	3	1	3	1	2	4	2	1	1	4	1	4	4	3	2	3	2	2	1	2	3	2	2	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	1	3	1	2	1	3	2	3	3	1	1	2	3	4	1	4	4	1	4	1	2	4	4	1	1	1	3	3	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	2	4	4	2	2	1	2	2	3	3	3	3	3	1	3	2	1	1	3	3	3	2	1	4	4	1	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	4	2	4	2	1	2	2	3	2	3	2	2	4	2	3	3	4	2	1	2	3	1	2	1	1	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	3	1	3	4	3	1	2	1	3	4	3	4	1	2	1	2	3	1	2	3	2	2	3	3	4	1	1	2	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	4	3	1	2	4	2	4	2	2	1	3	2	1	3	1	2	2	4	2	3										

HINT - SHEET**SUBJECT : PHYSICS****SECTION-A**1. **Ans (4)**

Width of central bright fringe

$$= \frac{2D\lambda}{a} = \frac{2 \times 2 \times 6 \times 10^{-7}}{1 \times 10^{-3}}$$

$$= 2.4 \times 10^{-3} \text{ m} = 2.4 \text{ mm}$$

2. **Ans (4)**

$$R.P. \propto \frac{1}{\lambda}$$

3. **Ans (4)**

$$I = \frac{I_0}{2} \cos^2 60$$

5. **Ans (3)**

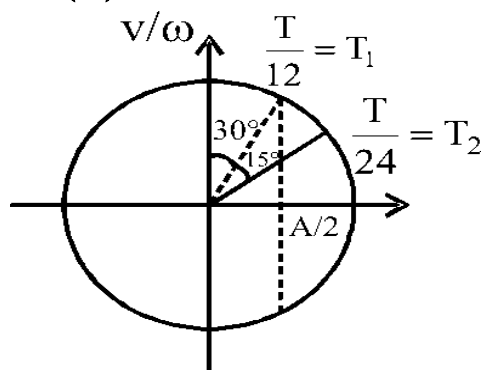
$$\text{From Brewster's law } \theta_p = \tan^{-1} \frac{\mu_2}{\mu_1}$$

$$\text{Here } \mu_1 = 1, \theta_p = 60^\circ, \mu_2 = \mu$$

$$\theta_p = \tan^{-1} \mu = 60^\circ$$

$$\mu = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow v = \frac{c}{\mu} = \frac{c}{\sqrt{3}} = \frac{3 \times 10^8}{\sqrt{3}} = \sqrt{3} \times 10^8 \text{ m/sec}$$

6. **Ans (4)**

NOAH STUDY HUB



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TG: @Chalnaayaaar

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- ALLEN MODULES 2023-24 ✓
- ALLEN EXTRA EDGE ✓
- ALLEN NCERT BASED OBJECTIVE ✓
- ALLEN SIKAR NEETriumph ✓
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(Latest + Previous Years) ⚡

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8. **Ans (3)**

$$m_T = -n = \frac{-(-f)}{u - (-f)}$$

9. **Ans (2)**

For an object placed at infinity the image after first refraction will be formed at v_1

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{+R} \quad \dots (i)$$

The image after second refraction will be found at v_2

$$\frac{\mu_3}{v_2} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{+R} \quad \dots (ii)$$

adding (i) and (ii)

$$\frac{\mu_3}{v_2} = \frac{\mu_3 - \mu_1}{R} \Rightarrow v_2 = \frac{\mu_3 R}{\mu_3 - \mu_1}$$

Therefore focal length will be $\frac{\mu_3 R}{\mu_3 - \mu_1} = 2R$

10. **Ans (3)**

$$\mu = \frac{1}{\sin \theta_C} = \frac{1}{\sin 45^\circ}$$

$$\mu = \sqrt{2}$$

13. **Ans (3)**

Rotation of reflected ray when mirror is rotated with θ , keeping incident ray fixed is $= 2\theta$

$$\text{So } 2 \times 20^\circ = 40^\circ$$

14. **Ans (1)**

$$n_1 \sin 45^\circ = n_3 \sin \theta \Rightarrow (1) \left(\frac{1}{\sqrt{2}} \right) = \sin \theta$$

$$\Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

15. **Ans (4)**

$$d_{app} = \frac{d}{\mu} = \frac{12}{4/3} = 9 \text{ cm}$$

18. **Ans (4)**

at $t = 1 \text{ sec.}$

$$y_1 = a \sin \left(\frac{\pi}{2} + \phi \right), y_2 = b \sin \left(\frac{2\pi}{3} + \phi \right)$$

$$\Delta \phi = \left(\frac{2\pi}{3} + \phi \right) - \left(\frac{\pi}{2} + \phi \right) \Rightarrow \Delta \phi = \frac{\pi}{6}$$

19. **Ans (3)**

$$\frac{3}{4} (\text{T.E.}) = \text{K.E.} \quad |T = 2 \text{ sec.}$$

$$\frac{3}{4} \left(\frac{1}{2} kA^2 \right) = \frac{1}{2} kA^2 \cos^2 \omega t$$

$$\frac{\sqrt{3}}{2} = \cos \omega t$$

$$\frac{\pi}{6} = \omega t \Rightarrow \frac{\pi}{6} = \frac{2\pi}{T} t$$

$$t = \frac{T}{12} = \frac{2}{12} = \frac{1}{6} \text{ sec.}$$

20. **Ans (3)**

$$V_{\max} = A\omega$$

$$4.4 = (7 \times 10^{-3})\omega$$

$$\omega = \frac{4.4}{7 \times 10^{-3}} \Rightarrow \frac{2\pi}{T} = \frac{4.4}{7 \times 10^{-3}}$$

$$T = 0.01 \text{ sec}$$

22. **Ans (4)**

$$V = -10 (2\pi) \sin \left(2\pi t + \frac{\pi}{2} \right)$$

$$\text{at } t = \frac{1}{6}$$

$$V = -20\pi \sin \left(\frac{\pi}{3} + 90^\circ \right) = -20\pi \cos \frac{\pi}{3}$$

$$V = -20 \left(\frac{1}{2} \right) = 10\pi$$

$$V = 31.4 \text{ cm/s}$$

SECTION-B

38. **Ans (3)**

First body is $\frac{\pi}{6}$ ahead of mean position and second body is $\frac{\pi}{6}$ behind of mean position, therefore the phase difference is $\frac{\pi}{3}$.

39. **Ans (1)**

For viewing far objects, concave lenses are used and for concave lens

$$u = \text{wants to see} = -60 \text{ cm}; v = \text{can see} = -15 \text{ cm}$$

$$\text{so from } \frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow f = -20 \text{ cm.}$$

40. **Ans (2)**

$$m = \frac{f_0}{f_e}$$

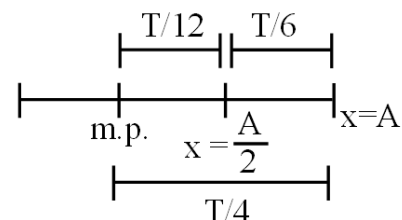
$$L = f_0 + f_e$$

41. **Ans (4)**

$$\delta = 180^\circ - 60^\circ = 120^\circ$$

43. **Ans (1)**

$$\frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2} = \frac{2}{3}$$

because f_1 and f_2 will be of opposite nature.44. **Ans (1)**

$$\frac{T}{4} - \frac{T}{12} = \frac{T}{6}$$

SUBJECT : CHEMISTRY**SECTION-A**51. **Ans (3)**

$a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ \rightarrow$ is the parameter for crystal triclinic.

52. **Ans (2)**

$$\frac{r^+}{r^-} = \frac{95}{181} = 0.52 \text{ (this value is present in between normal range so, C.N. remain 6)}$$

53. **Ans (2)**

The empty space in figure is octahedral void because it is surrounded by six spheres.

64. **Ans (1)**

$$k = \frac{2.303}{t} \log \frac{C_0}{C_t}$$

$$\frac{t_{0.999}}{t_{0.5}} = \frac{\frac{2.303}{k} \log \frac{100}{0.1}}{\frac{2.303}{k} \log \frac{100}{50}} = \frac{3}{0.3} = \frac{10}{1} = 10$$

81. **Ans (1)**NCERT - 12th**SECTION-B**86. **Ans (1)**In bcc $\sqrt{3}a = 4r$

$$\frac{\sqrt{3}}{2} a = 2r$$

$$\therefore 2r = \frac{\sqrt{3}}{2} \times 4 \times 10^{-10} = 3.46 \times 10^{-10} \text{ m}$$

87. **Ans (1)**

Total No. of octahedral void per atom in

$$\text{ccp} = \frac{4}{4} = 1$$

91. **Ans (3)**

$$[A]_t = [A] - Kt = 1 - 0.001 \times 10 \times 60$$

$$[A]_t = 0.4 \text{ M}$$

$$[B]_t = 0.001 \times 10 \times 60 = 0.6 \text{ M}$$

96. **Ans (2)**

M - C bond becomes stronger due to synergic bonding.

SUBJECT : BIOLOGY-I**SECTION-A**106. **Ans (1)**

NCERT XII, Pg # 52, 53

108. **Ans (2)**

NCERT (XIIth) Pg. # 47 (E) 51 (H)

113. **Ans (3)**

NCERT (XII) (E) Pg. # 45 Para# 3.2

114. **Ans (2)**

NCERT (XII) (E) Pg. # 46 Para# 3.2

115. **Ans (1)**

NCERT (XII) (E) Pg. # 47 Para# 3.2

117. **Ans (4)**

NCERT XIIth Pg # 60

118. **Ans (1)**

NCERT (XIIth) Pg. # 60

122. **Ans (3)**

NCERT-XII, Pg. # 233

123. **Ans (1)**

NCERT XII Pg. # 236

126. **Ans (4)**

Module Pg.# 17

129. **Ans (2)**

Allen Module page no. 26

132. **Ans (2)**

NCERT Pg.# 249

135. **Ans (2)**

NCERT XII, Pg. # 249

SECTION-B140. **Ans (4)**

NCERT Pg. # 51(E), 55(H)

143. **Ans (2)**

NCERT (XII) (E) Pg. # 48 Para# 3.3

144. **Ans (3)**

NCERT (XIIth) Pg. # 60 (E) 66 (H)

SUBJECT : BIOLOGY-II**SECTION-A**159. **Ans (1)**

NCERT-XII, Pg No.# 223

162. **Ans (3)**

NCERT Pg # 98

165. **Ans (2)**NCERT (XIIth) Pg. # 115166. **Ans (1)**

NCERT Reference:

Biology -XII, Page No.-110

169. **Ans (1)**

NCERT (XII) Pg.# 116

170. **Ans (2)**

NCERT (XII) Pg. # 117

176. **Ans (4)**

NCERT (XII) Pg. # 119

177. **Ans (1)**

NCERT Pg#112

178. **Ans (1)**

NCERT-XII, Pg. # 116

182. **Ans (3)**

NCERT-XII, Pg. # 116

185. **Ans (4)**

NCERT-XII, Pg # 121

SECTION-B196. **Ans (2)**

Module Pg. # 330

198. **Ans (4)**

NCERT-XII Pg # 117

200. **Ans (3)**

NCERT-XII, Pg. # 112